Jisang Lee

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Education

Hanbat National University (Advisor: Prof. Haneol Jang) *Daejeon*

M.S. Student in Computer Engineering Mar. 2025 - Present

GPA 4.5/4.5

Hanbat National University (Advisor: Prof. Haneol Jang) *Daejeon*

B.S. in Computer Engineering Mar. 2019 - Feb. 2025

GPA 4.09/4.5 (Major: 4.28/4.5)

Publications

Conference

[1] **J.S. Lee**†, S.Y. Oh†, C. Park, and H. Jang, "Uncertainty-Guided Structural Routing and Consensus-Aware Re-Ranking for Open-Vocabulary Camouflaged Object Segmentation", *Asian Conference on Computer Vision* **(ACCV, BK21 CS Outstanding Conference)**, 2026. To appear.

[2] **J.S. Lee**, H.W. Seo, C. Park, and H. Jang, "Crop-Paste Technique for Enhancing Object Detection Performance in Synthetic Aperture Radar Images", *Korea Computer Congress* (*KCC*), 2024.

[3] H.W. Seo, **J.S. Lee**, C. Park, and H. Jang, "Data Preprocessing and Augmentation Methods for SAR to Optical Image Translation", *Korea Computer Congress* (*KCC*), 2024.

[4] H.W. Seo, **J.S. Lee**, D.S. Kim, S.Y. Oh, and H. Jang, "Effective Detection of Generated Images Using Fast Fourier Transform and Discrete Cosine Transform", *Korean Digital Forensics Society* (*KDFS*) *Winter Conference*, 2023.

[5] **J.S. Lee**, D.S. Kim, S.Y. Oh, and H. Jang, "Lightweight Fingerprint Forgery Detection Technique Utilizing Median Filter and Ensemble Model", *Korean Digital Forensics Society* (*KDFS*) *Summer Conference*, 2023.

[6] D.S. Kim, **J.S. Lee**, S.Y. Oh, and H. Jang, "Effective Data Augmentations for Fingerprint Recognition", *Korean Digital Forensics Society* (*KDFS*) *Summer Conference*, 2023.

[7] S.Y. Oh, **J.S. Lee**, D.S. Kim, and H. Jang, "Lightweight Deep Convolutional Neural Networks for Fine-Grained Leaf Classification", *Joint Conference on Communications and Information* (*JCCI*), 2023.

[8] E.J. Jang, **J.S. Lee**, D.S. Kim, S.Y. Oh, and H. Jang, "Development of Optical Character Recognition Model for Document Leak Investigation", *Korean Digital Forensics Society* (*KDFS*) *Winter Conference*, 2022.

†Equal contribution.

Journal

[1] **J.S. Lee**, H.W. Seo, E.K. Kim, and H. Jang, "HFGF-DINOv3: High-Frequency Guided Fusion and Phase-Aware Loss for Camouflaged Object Detection", **Pattern Recognition(SCIE, IF 9.1, Top JCR 6.9%)**, vol. 183, Part B, Art. 115066, Mar. 2027.

[2] J.H. Park, **J.S. Lee**, S.M. Pyo, and H. Jang, "Enhanced Copy-Paste Data Augmentation Using SAM for Precise Ship Detection", *Journal of IKEEE*, 2025.

[3] **J.S. Lee**, H.W. Seo, W.J. Park, H. Choi, K. Kim, E.K. Kim, and H. Jang, "Crop-Paste Data Augmentation Technique for Enhancing Object Detection Performance in Synthetic Aperture Radar Images", *Journal of IKEEE*, 2025.

[4] H.W. Seo, **J.S. Lee**, W.J. Park, H. Choi, K. Kim, and H. Jang, "Effective Data Preprocessing and Augmentation Methods of SAR Images for Optical Image Translation", *Journal of Broadcast Engineering*, 2025.

[5] H.W. Seo, D.S. Kim, S.Y. Oh, **J.S. Lee**, and H. Jang, "Effective Detection of Generated Images Using Frequency Transform", *Journal of KIISE* (*JOK*), 2025.

[6] **J.S. Lee**, H. Kim, J. Koo, H. Choi, and D.Y. Jeong, "Parcel-Based Crop Type Classification in UAV Imagery with SAM for Smallholder Farms", *Korean Journal of Remote Sensing*, 2024.

[7] D.S. Kim, S.Y. Oh, **J.S. Lee**, and H. Jang, "Effective Data Augmentations for Cross-Domain Fingerprint Recognition", *Journal of Digital Forensics*, 2024.

[8] **J.S. Lee**, D.S. Kim, S.Y. Oh, and H. Jang, "Lightweight Technique for Forged Fingerprint Detection Using Ensemble Models and Median Filters", *Journal of Digital Forensics*, 2023.

Research Experience

AIM Lab., Hanbat National University *Daejeon, Republic of Korea*

M.S. Researcher (Advisor: Prof. Haneol Jang) Mar. 2025 - Present

- Camouflaged and open-vocabulary segmentation with vision foundation and vision-language models; multispectral image understanding

AIM Lab., Hanbat National University *Daejeon, Republic of Korea*

Undergraduate Researcher (Advisor: Prof. Haneol Jang) May 2022 - Feb. 2025

- Conducted research on computer vision and deep learning

SI Analytics *Daejeon, Republic of Korea*

Research Intern Jul. 2024 - Sep. 2024

- Improved SAM-based parcel segmentation for UAV crop classification; published in KJRS (2024)

Research Projects

Open-Vocabulary Zero-Shot Detection for Military Objects

Collaborative Research with DataMaker Jan. 2025 - Oct. 2025

- Studied zero-shot military vehicle detection with vision-language models and evaluated recognition of unseen categories

SAR Image Analysis and Semantic Segmentation

Capstone Design with SkyPixel Jan. 2024 - Dec. 2024

- Studied SAR despeckling, scaling, and generative data augmentation; applied knowledge distillation to segmentation

X-ray Diagnosis of Neonatal Respiratory Distress Syndrome

Chungnam National University Hospital Jan. 2023 - Jul. 2023

- Developed neonatal X-ray lung segmentation and classification pipelines, with Grad-CAM for interpretation

Awards and Honors

Excellence Award	National Park Satellite Monitoring AI Challenge (**2nd/81 teams**)	*2026*
Gold Prize	Capstone Design Exhibition (SkyPixel), Hanbat National University	*2024*
Prize (3 awards)	Computer Engineering Portfolio Competition, Hanbat National University	*2022–2024*
Grand Prize	6th AI SPARK Challenge (**1st/141 teams**); Innopolis Foundation Chairman's Award	*2024*
Best Paper Award	KDFS Winter Conference; Society President's Award	*2023*
Special Prize	SW-Centered University Joint AI Competition (17th/227 teams); DEEPNOID Award	*2023*
Best Paper Award	KDFS Summer Conference; Korea Coast Guard Commissioner General Award	*2023*
5th/542 teams	Fake or Real: AI-Generated Image Detection Challenge, AI Connect	*2023*
Best Paper Award	KDFS Winter Conference; Society President's Award	*2022*
Special Prize	Creative Design Online Competition; Nanum Tech President's Award	*2022*
Encouragement Prize	SW-Centered University Joint AI Competition, Finals (8th/77 teams); SW-Centered University Council President's Award	*2022*
1st/192 teams	SW-Centered University Joint AI Competition, Preliminaries	*2022*

Technical Skills

Programming	Python, Java, C, C++
Libraries	PyTorch, NumPy, Pandas, scikit-learn
Languages	Korean (native), English (beginner)
Certifications	Deep Learning Specialization, Coursera (five courses), 2023
	Fingerprint Liveness Detection Competition (LivDet), Challenge 1, 2023

Teaching Experiences

Hanbat National University

Teaching Assistant (Prof. Haneol Jang) 2023 - 2026

- **Computer Vision** (graduate): Fall 2026 (ongoing); Spring 2026
- **Computer Vision** (4th-year undergraduate): Fall 2025
- **Artificial Intelligence and Applications** (3rd-year undergraduate): Spring 2025; Spring 2024
- **Data Structure** (2nd-year undergraduate): Spring 2023
- **Hanbat AI Intensive Course**: Jan. 2023

Academic Service

IEEE Transactions on Multimedia (TMM)

Reviewer 2026